Absorbing a Positivity Square
Fourteen Integers, Degree 88, 119 Gates

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1 Absorbing a positivity square: fourteen integers, degree 88

The polynomial below uses fourteen unrestricted integer variables, has exact ordinary total degree 88, and a division-free evaluator with 119 binary operations (67 multiplications, 27 additions and 25 subtractions). It is equivalent by explicit two-way maps on every zero to the pinned fifteen-variable free-scale degree-44 source. Consequently its zero nonexistence is equivalent in ordinary ZFC to syntactic Con(ZFC) under that unchanged compiler/calibration theorem. This is neither a proof of consistency nor a new end-to-end formal-kernel certificate.

1.1 The complete fixed equation

All unknowns are unrestricted integers:

$$j, C, z, h, r, v, w, T, c, n, f, y, p_2, p_3.$$

The six fixed numerals $\beta, K_0, M_C, M_F, A_c, B_c$ are exactly the authenticated source34 payloads, with M_F already shifted. The remaining names are polynomial abbreviations, not extra witnesses:

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = v, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = E(Yk), \\
I = (2r + 1)c^2, \quad Q = \Delta(f^2 - 1), \quad V = f^2(cT - R) - c, \\
M = 2q^3, \quad U = MH, \quad K_{\mathrm{lo}} = X(2X - 3c) - 2, \\
Z = Un + MK_{\mathrm{lo}} + H(Y - q^3), \\
D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B_{\mathrm{ud}} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2\big(4(Y\eta)^2 + (kq^3)^2\big), \\
G = B_{\mathrm{ud}} - 1 - p_2^2 - p_3(p_3 + 1).
\end{gathered}$$

Set

$$\begin{aligned}
C_1 &= (Z + Uk)^2 + 4U(LZ - U), \\
B &= f^2 - \Delta I^2 - 1, \\
C_2 &= V^2 - (Q - 1)y(2V + y) - 1, \\
N &= (C_1 - U^2G)^2 - 16U^4G\big(1 + L(L + k)\big).
\end{aligned}$$

The single equation is

$$\boxed{P = (N+1)(B+1)(C_2+1) - 1 = 0.}$$

The evaluator shares U^2G, q^3 and the already-counted registers. No division, radical, inequality or variable exponent appears in its literal DAG.

1.2 An exact, unsquared conjunction

For integers A, B with $A, B \neq -2$, the equation $(A+1)(B+1)(C+1) = 1$ is equivalent to $A = B = C = 0$: all three factors must be units, and the first two cannot be -1. Here N is a square modulo 4, hence $N \neq -2$. Write $A_0 = a+2$, so $\Delta = A_0^2 - 1$. If A_0 is odd, $B \equiv f^2 - 1 \pmod 4$; if A_0 is even, $B \equiv f^2 + I^2 - 1 \pmod 4$. Neither can be 2 (mod 4), so $B \neq -2$. Thus $P = 0$ is exactly $N = B = C_2 = 0$, with no nonnegativity assumption on N. For comparison, the general construction $\prod_i (3R_i + 1) = 1$ also encodes $R_i = 0$ over integers; the present residue exclusions avoid those extra scalar multiplications. No novelty is claimed for that general unit-product device.

1.3 Separation and extraction before invoking the source

Since $\beta = 2^{1064921} - 1$ is odd and greater than one, q is even and nonzero. Also H is odd and nonzero. Thus $U \neq 0$ on every integer assignment, and $|X| = |qW| \geq 4|q| \geq 8$. Introduce proof-only quantities

$$D = XY^2, \quad A = 4\big(1 + D(D+1)k^2\big), \quad b = 2L + k, \quad t = Z/U, \quad \rho = t + b.$$

The exact norm identity is

$$N/U^4 = (\rho^2 - A - G)^2 - 4AG. \tag{1}$$

Because $D(D+1) \geq 0$ for integer D, $A \geq 4$. At $N = 0$, (1) forces $G \geq 0$. Therefore $B_{\mathrm{ud}} = G + 1 + p_2^2 + p_3(p_3+1) > 0$, which excludes $Y = 0$ and $k = 0$. Directly, $B_{\mathrm{ud}} \leq Y^2 k^2 q^2$. Since $|D| \geq 8$, one has $D(D+1) \geq D^2/2$, and hence

$$A \geq 4 + 2X^2Y^4k^2 \geq 4 + 32q^2Y^4k^2 > Y^2k^2q^2 \geq B_{\mathrm{ud}} > G. \tag{2}$$

In particular $\rho \neq 0$, since (1) with $\rho = 0$ would force $A = G$. The rational numbers

$$p = \frac{\rho^2 - A + G}{2\rho}, \qquad u = \rho - p$$

satisfy $p^2 = G$ and $u^2 = A$. A rational square root of an integer is integral, so $p, u \in \mathbb{Z}$, and u is even. It follows that

$$t_{\mathrm{old}} = t - p = u - b \in \mathbb{Z}, \qquad t_{\mathrm{old}}^2 + 2(2L+k)t_{\mathrm{old}} + k^2 - 4 = 0. \tag{3}$$

The argument explicitly handles equal square classes: for two roots, the separation $A > G$ rules out the sole cancelling-sum obstruction. It does not claim that pairwise distinctness suffices for a norm with more than two roots.

1.4 Complete maps in both directions

At any new zero, retain all shared coordinates and set

$$p_{1,\mathrm{old}} = p, \qquad n_{\mathrm{old}} = n - p.$$

Since Z is affine in n with coefficient U, (3) makes the old first carrier zero. Its strong and auxiliary residuals are unchanged, and its guard is zero because $B_{\mathrm{ud}} = p^2 + 1 + p_2^2 + p_3(p_3+1)$. This is a literal

zero of the pinned fifteen-variable polynomial, so its complete ordinary soundness theorem now applies. No premature CRT or canonical assumption has been used.

Conversely, from any old zero, delete p_1 and set $n_{\text{new}} = n_{\text{old}} + p_1$. Then $G = p_1^2$ and $Z_{\text{new}} = Z_{\text{old}} + Up_1$. The universal identity

$$N_{\text{new}} = C_{1,\text{old}}\left[C_{1,\text{old}} + 4Up_1(Z_{\text{old}} + Ub) + 4U^2p_1^2\right]$$

gives $N_{\text{new}} = 0$, with the other two residuals unchanged. This applies, in particular, to every member of the authentic canonical source34 family supplied from a finite ZFC refutation. No new rank-tuning or two-square representation theorem is required.

1.5 Exact coefficient, cost and boundary

The degrees of C_1, B, C_2, G, N are $22, 12, 16, 18, 60$. For the leading homogeneous forms, write

$$q_0 = 6\beta j, \quad X_0 = 12q_0w, \quad Y_0 = v, \quad R_0 = q_0^4z - 12q_0^3wC,$$
$$a_0 = X_0Y_0, \quad U_0 = 8q_0^3X_0Y_0.$$

Then

$$N_{60} = U_0^4R_0^4q_0^8(q_0^8 + 16X_0^2Y_0^4),$$
$$B_{12} = -4a_0^2r^2c^4, \qquad (C_2)_{16} = 2a_0^2f^4R_0y.$$

Their product is the unique degree-88 part of P. In particular,

$$[j^{56}w^8v^8z^5r^2c^4f^4y]P = -2^{87}3^{64}\beta^{56} \neq 0.$$

Thus the degree is attained on the actual fixed coefficients. The 119-gate count includes all scalar multiplications and squares. This lowers the historical fourteen-variable degree-108/119-gate endpoint's degree at unchanged evaluator cost; it is a different tradeoff from the fifteen-variable degree-44/111-gate source.

The independent audit checks all 33 named registers, literal liveness and gate counts, the nonzero coefficient, complete coefficient bytes, source pins, identities, residue cases, signed bounds, and deliberate gate mutations. Normal and optimized runs pass. These finite checks do not replace the unbounded source34 compiler/refutation theorem, whose source closure remains an explicit dependency.